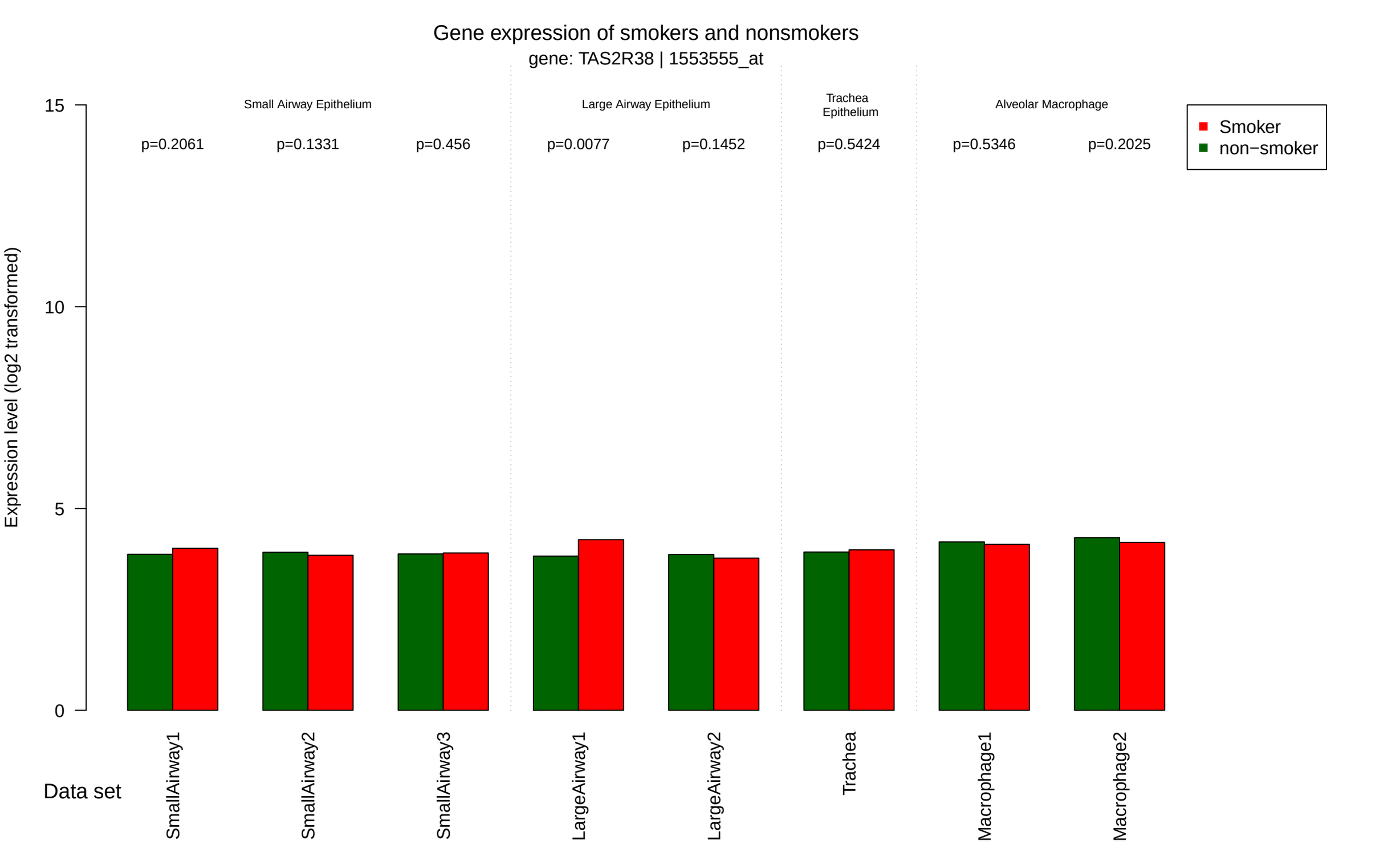
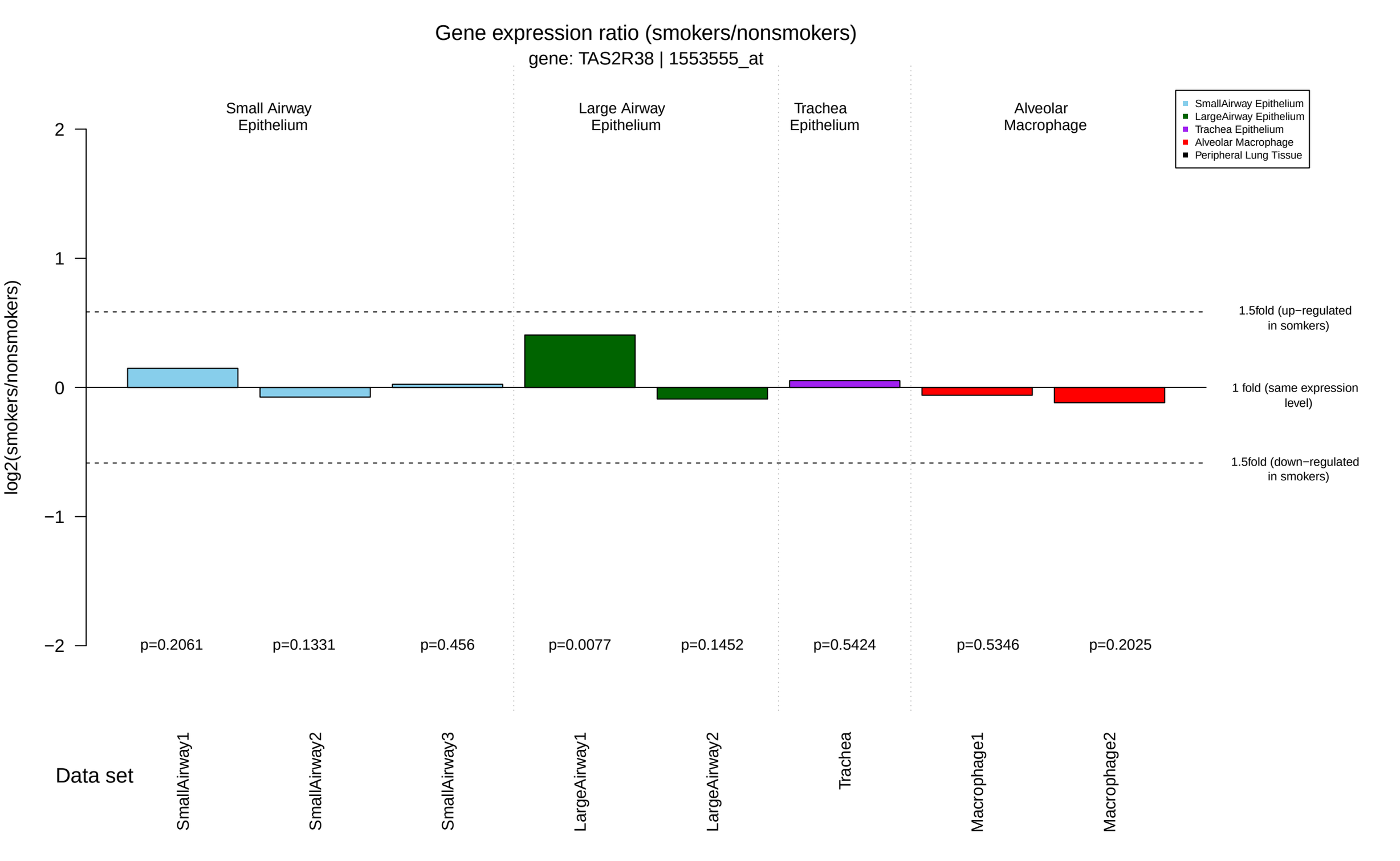
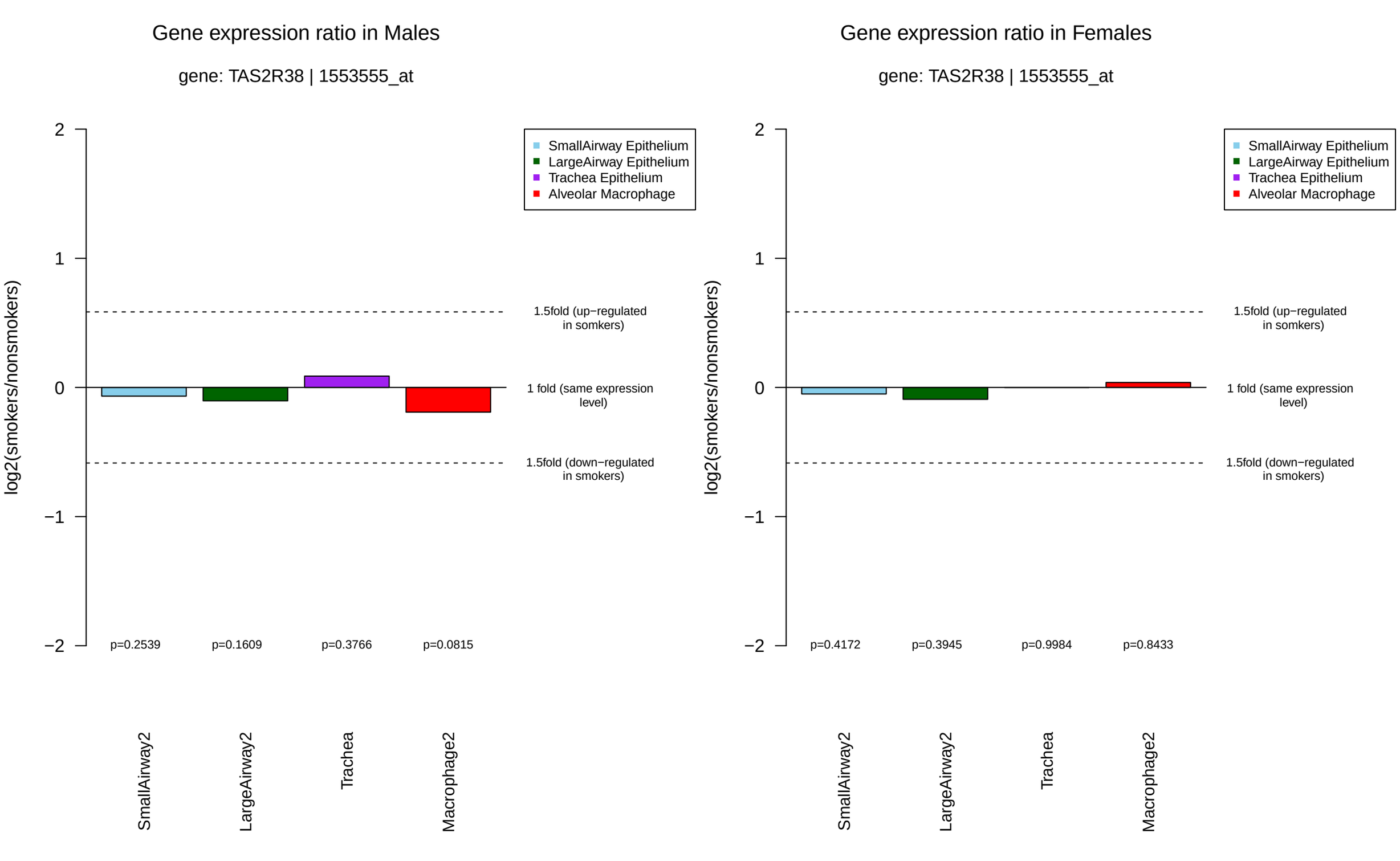
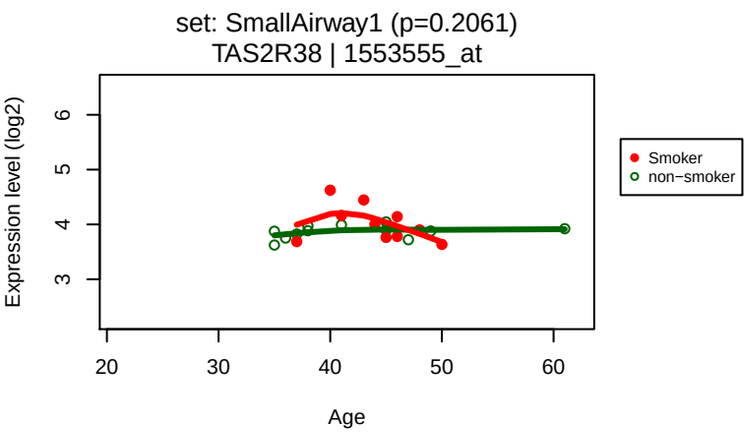
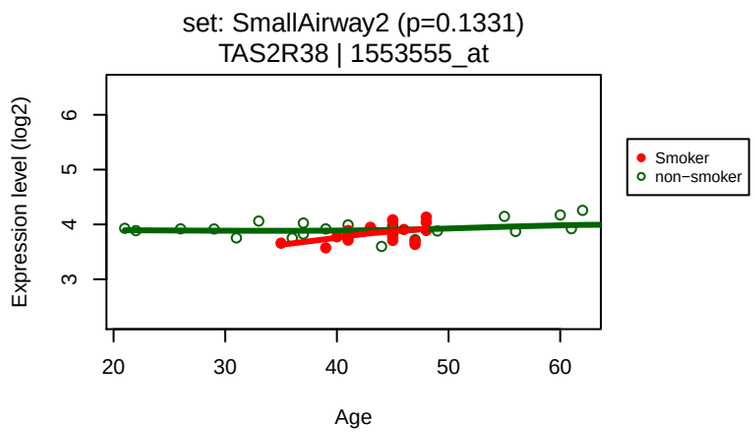
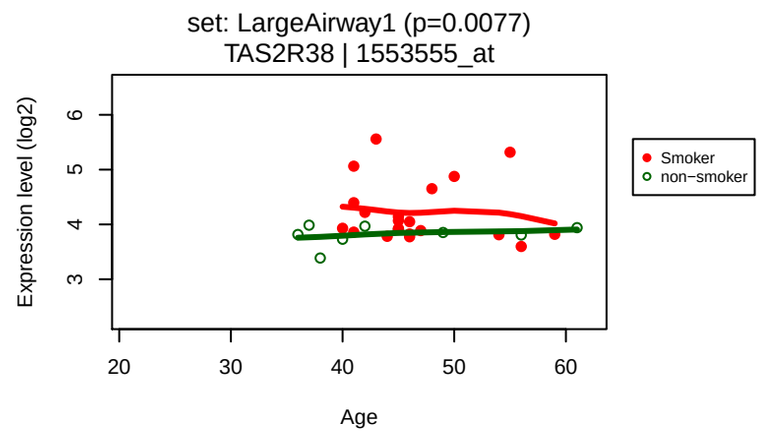

Gene expression of smokers and nonsmokers
gene: TAS2R38 | 1553555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2061
p=0.1331
p=0.456
p=0.0077
p=0.1452
p=0.5424
p=0.5346
p=0.2025
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

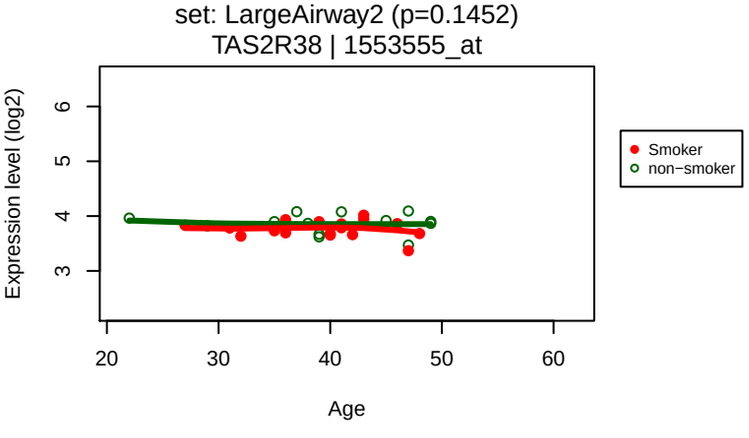

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R38 | 1553555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2061
p=0.1331
p=0.456
p=0.0077
p=0.1452
p=0.5424
p=0.5346
p=0.2025
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

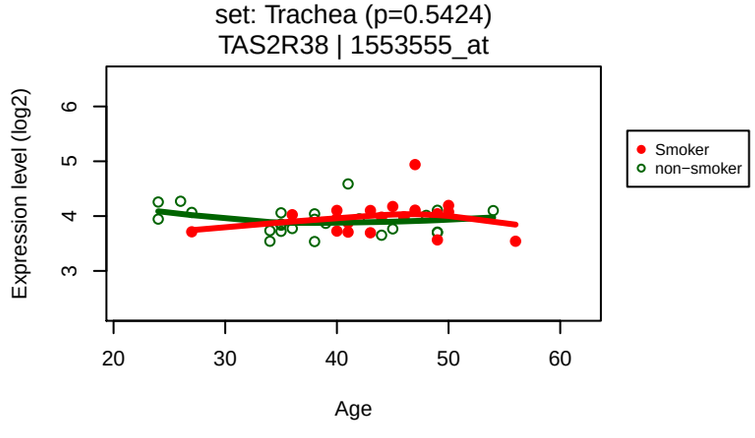

Gene expression ratio in Males
gene: TAS2R38 | 1553555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2539
p=0.1609
p=0.3766
p=0.0815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R38 | 1553555_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4172
p=0.3945
p=0.9984
p=0.8433
SmallAirway2
LargeAirway2
Trachea
Macrophage2

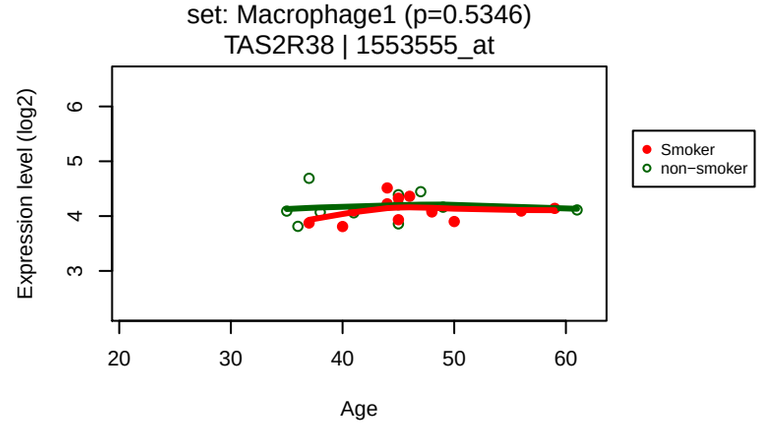

set: SmallAirway1 (p=0.2061)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker

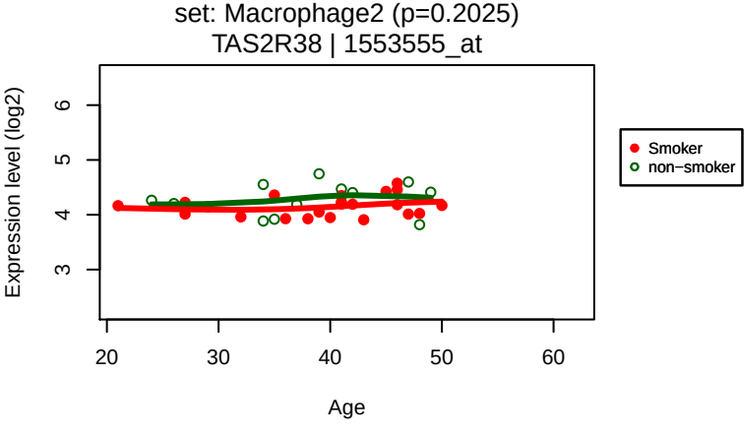

set: SmallAirway2 (p=0.1331)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker

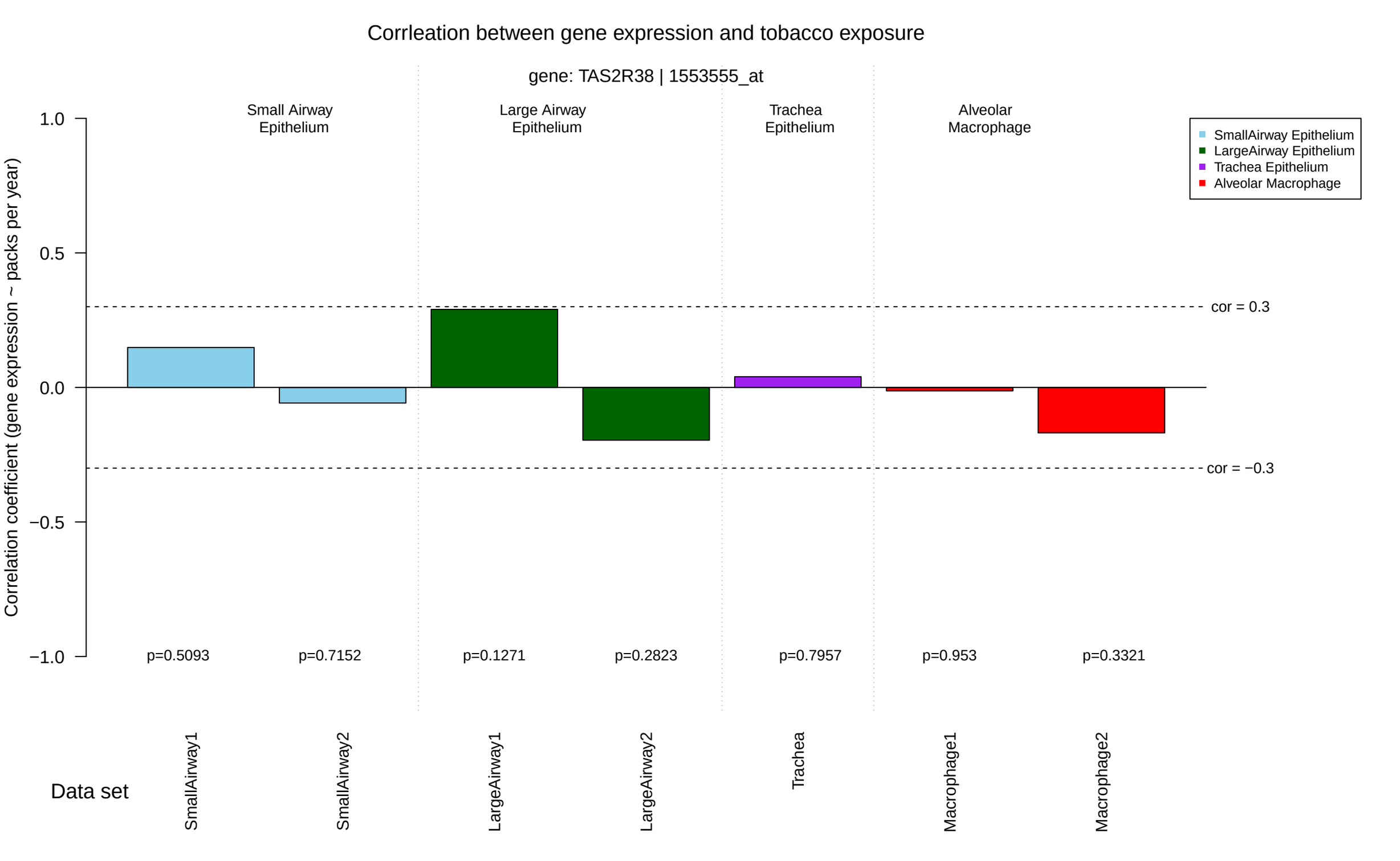

set: LargeAirway1 (p=0.0077)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1452)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5424)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5346)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2025)
TAS2R38 | 1553555_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R38 | 1553555_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5093
p=0.7152
p=0.1271
p=0.2823
p=0.7957
p=0.953
p=0.3321
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2